

SUBHANKER KUMAR

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Education

School of Information Technology, RGPV

Bachelor of Technology – Computer Science & Engineering (AI/ML)

Bhopal, MP, India

2023 – 2027

Delhi Public School

Intermediate – Physics, Chemistry, Mathematics

Gaya, Bihar, India

2020 – 2022

Technical Skills

- **Languages:** C, C++, Python
- **Core CS:** DSA, OOP, CN, DBMS, OS
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Streamlit
- **Machine Learning:** Supervised Learning, Isolation Forest, PCA, Feature Engineering, Anomaly Detection, Deep Learning, ANN, CNN, RNN, LSTM, NLTK, TF-IDF
- **AI:** LLM, Prompt Chaining, RAG, Agentic AI
- **Databases:** MySQL
- **Tools & Platforms:** Git, GitHub, VS Code

Projects

Zero-Day Anomaly Detection Intrusion Detection System Python, Scikit-learn, Streamlit, Plotly, PCA, Isolation Forest

- Developed an anomaly-based Intrusion Detection System using the Isolation Forest algorithm to detect zero-day cyber attacks on the NSL-KDD dataset without requiring labeled training data.
- Built an end-to-end machine learning pipeline incorporating data preprocessing, Mutual Information-based feature selection, PCA for dimensionality reduction, and model evaluation, achieving 97.8% accuracy and 94.8% F1-score.
- Designed an interactive Streamlit dashboard with Plotly visualizations for anomaly monitoring, performance analysis, feature insights, and real-time network traffic exploration.

Satellite Imagery Land-Use Classification using CNNs

Python, PyTorch, Torchvision, ResNet18, Scikit-learn, Grad-CAM, Streamlit

- Developed a deep learning pipeline to classify 27,000+ Sentinel-2 satellite images across 10 land-use/land-cover classes using CNN-based image classification.
- Implemented and evaluated 2 CNN architectures — a custom 3–4 block CNN trained from scratch and ResNet18 transfer learning pretrained on ImageNet.
- Applied data augmentation, normalization, stratified train/validation/test splitting, Adam optimization, and learning-rate scheduling to improve model generalization.
- Evaluated models using accuracy, precision, recall, F1-score, and 10-class confusion matrices, enabling quantitative comparison between baseline and transfer-learning approaches.
- Integrated Grad-CAM explainability to generate class-activation heatmaps identifying image regions influencing predictions and deployed the inference pipeline through a Streamlit web application.

DPI Engine — Offline Deep Packet Inspection System

 C++17 | libpcap | TCP/IP | TLS | Multithreading | PCAP

- Developed an offline DPI engine in C++17 for Ethernet PCAP analysis, supporting IPv4, TCP/UDP parsing, bidirectional flow tracking, and protocol-aware traffic inspection.
- Implemented TLS ClientHello SNI and HTTP Host extraction to classify 6+ common applications/domains including YouTube, Facebook, TikTok, GitHub, and Google.
- Built a configurable rule engine with 3 blocking policies — application, source IPv4, and domain — and generated filtered PCAP files by dropping matching traffic.
- Engineered both single-threaded and multithreaded processing pipelines using thread-safe queues, configurable worker threads, and modular components for packet parsing, flow management, protocol extraction, and rule evaluation.

Achievements

- Solved 250+ Data Structures and Algorithms problems on LeetCode, with strong proficiency in arrays, trees, graphs, dynamic programming, and backtracking.